Global Angular Blocks for the Middle-Ratio Shell Proxy

Global analytic angular module

Abstract

The global structural reduction for the spherical-shell proxy leaves an angular rounding defect $\mathcal{E}_{\mathrm{ang}}$ and a smooth positive payment $\mathcal{H}$. This note controls the defect for every $0 < \rho < 1$ and $K > 0$ with at least one active radial layer. A single paired radial-block argument, using only the sharp slope cap $-A' \leq 1/2$ and the exact strict angular rounding, gives

$$\mathcal{E}_{\mathrm{ang}} < \frac{(1-\rho^2)K^2}{6} - \frac{N}{2},$$

where N is the number of active shifted radial layers. On the upward-closed middle-ratio region $\rho_c \leq \rho \leq 39/50$, $K \geq k_{11}(\rho)$, one has $N \geq 1$. The global middle-ratio theorem is therefore reduced to the smooth scalar inequality (19), which is established in the companion scalar-positivity module.

1 Definitions and the target payment

For $a > 0$ and $z \geq 0$, define

$$G_a(z) = \begin{cases} \dfrac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos \dfrac{z}{a}\right), & 0 \leq z < a, \\ 0, & z \geq a. \end{cases} \tag{1}$$

Fix $0 < \rho < 1$ and $K > 0$, and put

$$A(z) := G_K(z) - G_{\rho K}(z), \qquad T := A(0) = \frac{(1-\rho)K}{\pi}. \tag{2}$$

The function A decreases strictly from T to 0. Denote its decreasing inverse by $R : [0, T] \to [0, K]$, and set

$$F(t) := R(t)^2, \qquad q(z) := -A'(z), \qquad \tau := A(\rho K) = K G_1(\rho). \tag{3}$$

Direct differentiation gives

$$q(z) = \frac{1}{\pi}\begin{cases} \arccos \dfrac{z}{K} - \arccos \dfrac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos \dfrac{z}{K}, & \rho K \leq z < K. \end{cases} \tag{4}$$

Consequently

$$0 \leq q(z) \leq \frac{1}{2}, \qquad q \text{ increases on } (0, \rho K), \qquad q \text{ decreases on } (\rho K, K). \tag{5}$$

Let

$$x_n := n - \frac{1}{4}, \qquad M(r) := \max\left\{0, \left\lceil r - \frac{1}{2} \right\rceil\right\}, \qquad N := \#\{n \geq 1 : x_n < T\}. \tag{6}$$

For $1 \leq n \leq N$, write

$$r_n := R(x_n), \qquad m_n := M(r_n), \tag{7}$$

and define the exact angular rounding defect

$$\mathcal{E}_{\mathrm{ang}}(\rho, K) := \sum_{n=1}^{N} (m_n^2 - r_n^2). \tag{8}$$

The strict convention is built into M: at $r = m + 1/2$ one has $M(r) = m$, not $m + 1$.

The smooth payment obtained from the retained radial Stieltjes remainder is

$$\begin{aligned} \mathcal{H}(\rho, K) := \frac{\rho^2 K^2}{4} &- \frac{\pi \rho K}{4(1-\rho)} \\ &+ \frac{2\pi\rho K}{\arccos \rho} \left(\frac{\tau}{4(1+2\tau)} + \frac{3}{32} \right) \\ &+ \int_{\rho K}^{K} 2z \, \frac{A(z)(1+A(z))}{(1+2A(z))^2} \, dz. \end{aligned} \tag{9}$$

The preceding structural reduction proves the direct lower bound

$$W(\rho, K) - P(\rho, K) \geq \mathcal{H}(\rho, K) - \mathcal{E}_{\mathrm{ang}}(\rho, K). \tag{10}$$

This note derives a smooth upper bound for $\mathcal{E}_{\mathrm{ang}}$.

2 A one-layer paired-block estimate

The cap $q \leq 1/2$ in (5) forces a useful separation: one unit of radial action spans at least two units of angular radius. The shift $x_n = n - 1/4$ leaves a left block of length $3/4$, enough to pay the worst angular rounding in that layer.

Lemma 1 (One-layer paired block)**.** *For every active n,*

$$r_n \leq \frac{4}{3} \int_{n-1}^{x_n} R(t)\, dt - \frac{3}{4}, \tag{11}$$

and

$$m_n^2 - r_n^2 < r_n + \frac{1}{4}. \tag{12}$$

Both statements remain valid, with the displayed strictness, at every radial–angular common wall.

Proof. Fix $t \in [n-1, x_n]$ and put $z = R(t) \geq r_n$. Since $A(r_n) = x_n$, (5) gives

$$x_n - t = A(r_n) - A(z) = \int_{r_n}^{z} q(s)\, ds \leq \frac{z - r_n}{2}.$$

Hence

$$R(t) \geq r_n + 2(x_n - t).$$

Integration over the interval of length $3/4$ yields

$$\int_{n-1}^{x_n} R(t)\,dt \ge \frac{3}{4}r_n + \frac{9}{16},$$

which is (11).

By strict counting, $m_n < r_n + 1/2$ even when r_n is a half-integer: at $r_n = m + 1/2$ the value is $m_n = m$. Squaring gives

$$m_n^2 < (r_n + 1/2)^2 = r_n^2 + r_n + 1/4,$$

and proves (12). □

Proposition 2 (Global angular block bound)**.** *For every $0 < \rho < 1$ and every $K > 0$ with $N \ge 1$,*

$$\boxed{\mathcal{E}_{\mathrm{ang}}(\rho, K) < \frac{(1-\rho^2)K^2}{6} - \frac{N}{2}.} \tag{13}$$

In particular,

$$\mathcal{E}_{\mathrm{ang}}(\rho, K) < \frac{(1-\rho^2)K^2}{6} - \frac{1}{2}. \tag{14}$$

Proof. The left blocks $[n-1, x_n]$, $1 \le n \le N$, are disjoint subsets of $[0, T]$. Summing (11) and then (12) gives

$$\begin{aligned}
\mathcal{E}_{\mathrm{ang}} &< \sum_{n=1}^{N}\left(r_n + \frac{1}{4}\right) \\
&\le \frac{4}{3}\sum_{n=1}^{N}\int_{n-1}^{x_n} R(t)\,dt - \frac{N}{2} \\
&\le \frac{4}{3}\int_0^T R(t)\,dt - \frac{N}{2}.
\end{aligned}$$

The area identity for decreasing inverses gives

$$\int_0^T R(t)\,dt = \int_0^K A(z)\,dz.$$

Scaling $z = as$ in (1) and using

$$\int_0^1 \sqrt{1-s^2}\,ds = \frac{\pi}{4}, \qquad \int_0^1 s\arccos s\,ds = \frac{\pi}{8},$$

shows that

$$\int_0^a G_a(z)\,dz = \frac{a^2}{8}.$$

Consequently

$$\int_0^T R(t)\,dt = \frac{(1-\rho^2)K^2}{8},$$

which proves (13). Equation (14) follows from $N \ge 1$. □

Combining the structural and angular estimates removes the staircase defect from the live handoff altogether.

Proposition 3 (Direct analytic proxy gap)**.** *For every* $0 < \rho < 1$ *and* $K > 0$ *with* $N \geq 1$*,*

$$\boxed{W(\rho, K) - P(\rho, K) > \mathcal{H}(\rho, K) - \frac{(1-\rho^2)K^2}{6} + \frac{N}{2}.} \tag{15}$$

This one inequality includes the first and terminal radial pieces and every strict angular wall; no separate common-jump estimate is a premise.

Proof. Insert (13) into (10). □

3 The upward-closed middle-ratio scalar inequality

Set

$$\rho_c := \frac{1}{1+2\pi}, \qquad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}, \tag{16}$$

and consider the upward-closed middle-ratio region

$$\mathcal{R}_{\mathrm{c}} := \left\{(\rho, K) : \rho_c \leq \rho \leq \frac{39}{50}, \quad K \geq k_{11}(\rho)\right\}. \tag{17}$$

On this set,

$$T \geq \frac{(1-\rho)k_{11}(\rho)}{\pi} = \sqrt{1 + \frac{132(1-\rho)^2}{\pi^2}} > 1, \tag{18}$$

so $N \geq 1$. Proposition 3 reduces the global middle-ratio theorem to the single inequality

$$\boxed{\mathcal{S}(\rho, K) := \mathcal{H}(\rho, K) - \frac{(1-\rho^2)K^2}{6} + \frac{1}{2} > 0} \qquad ((\rho, K) \in \mathcal{R}_{\mathrm{c}}). \tag{19}$$

Everything in $\mathcal{S}$ is smooth at every finite point of $\mathcal{R}_{\mathrm{c}}$; the floor functions and all radial–angular common jumps have disappeared. Substituting (9) gives the explicit form

$$\begin{aligned}\mathcal{S}(\rho, K) = {} & \frac{(5\rho^2-2)K^2}{12} - \frac{\pi\rho K}{4(1-\rho)} + \frac{1}{2} \\ & + \frac{2\pi\rho K}{\arccos\rho}\left(\frac{KG_1(\rho)}{4(1+2KG_1(\rho))} + \frac{3}{32}\right) \\ & + \int_{\rho K}^{K} 2z\,\frac{A(z)(1+A(z))}{(1+2A(z))^2}\,dz.\end{aligned} \tag{20}$$

The elementary identity

$$\frac{u(1+u)}{(1+2u)^2} = \frac{1}{4} - \frac{1}{4(1+2u)^2} \tag{21}$$

turns the same scalar gap into the more useful loss form

$$\begin{aligned}\mathcal{S}(\rho, K) = {} & \frac{(1+2\rho^2)K^2}{12} - \frac{1}{2}\int_{\rho K}^{K} \frac{z}{(1+2A(z))^2}\,dz \\ & - \frac{\pi\rho K}{4(1-\rho)} + \frac{1}{2} + \frac{2\pi\rho K}{\arccos\rho}\left(\frac{KG_1(\rho)}{4(1+2KG_1(\rho))} + \frac{3}{32}\right).\end{aligned} \tag{22}$$

Thus the only non-endpoint quantity still requiring control is the positive boundary-layer loss integral in (22).

Remark 4 (Status and next step). The direct proxy gap (15) is proved analytically from one retained-radial estimate and one one-layer angular estimate. The first remaining step within this module is the smooth scalar inequality (19). The companion scalar-positivity module proves it on all of $\mathcal{R}_{\mathrm{c}}$ by a uniform upper bound for the boundary-layer loss in (22) and strict positivity on the base curve. No upper cutoff in K occurs in the angular estimate or in the verification that $N \geq 1$.